Deliverable 3: Firmware Update Verification (Task 3)

1. Objective

To implement an automated verification pipeline that confirms firmware authenticity and integrity throughout the update lifecycle, including strict rollback protection mechanisms.

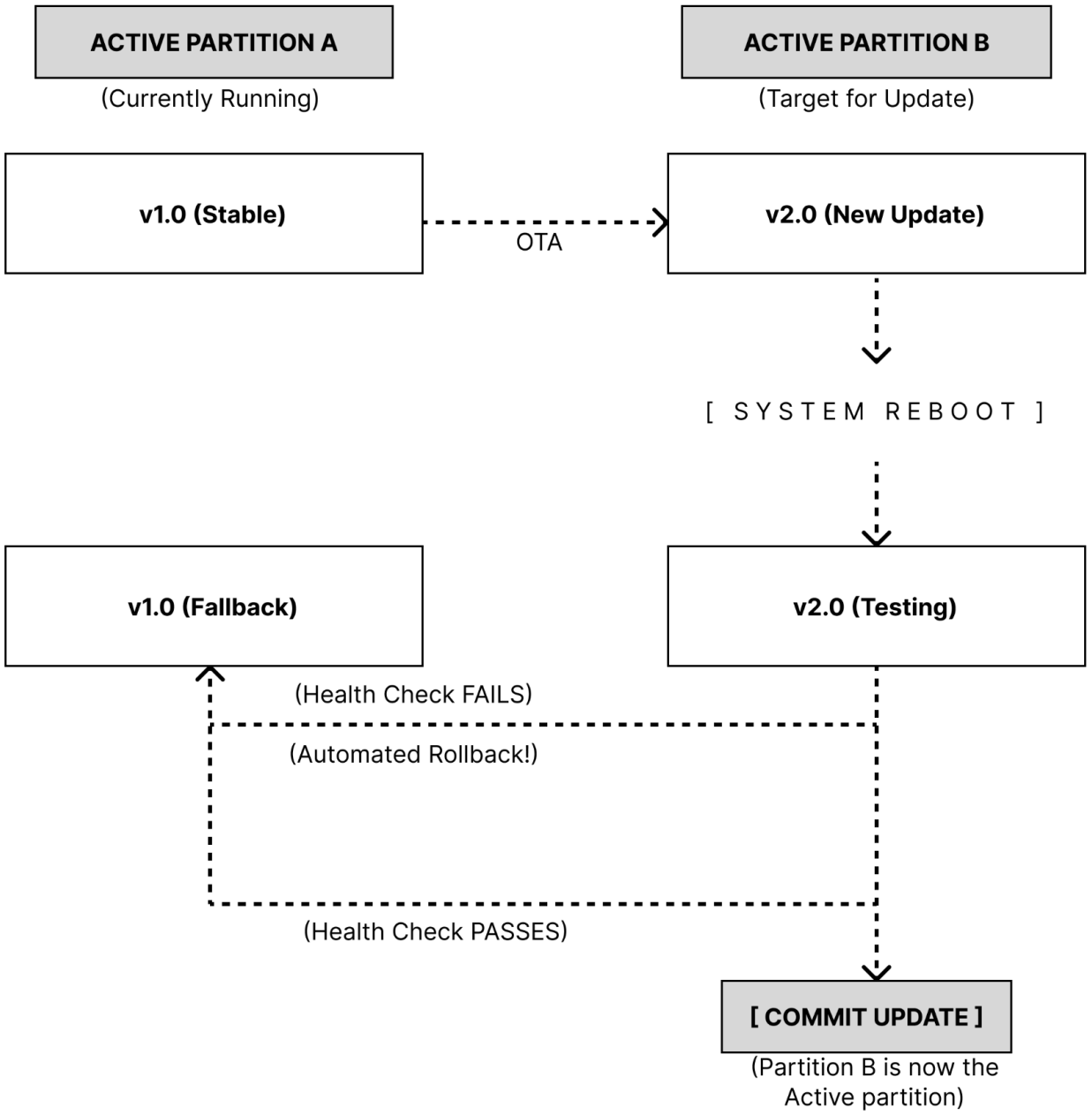
2. Verification Lifecycle

This blueprint defines the "Gatekeeper" logic executed by the Mender client on your virtual device.

FIRMWARE VERIFICATION GATEWAY	
PHASE	ACTION & VALIDATION
1, PRE-INSTALLATION	● Verify RSA-2048 Signature against Public Key ● Match Device Type (qemux86-64) ● Check SHA-256 Payload Hash
2. INSTALLATION	● Atomic Write to Passive Partition (A/B) ● Stream data with 100% Integrity Check
3. POST-INSTALLATION	● Verification Reboot into new feature ● “Heartbeat” Confirmation back to Mender Server ● Finalize Partition Swap (Commit Update)

3. Rollback Protection Mechanism

This illustrates how the system protects against "Downgrade Attacks" or corrupted updates.



4. Implementation Evidence

- **Signature Enforcement:** The device was configured to strictly require a valid RSA signature; any artifact built without the `-k` private key flag was rejected by the client.
- **Integrity Validation:** Artifact `custom-signed-v1.0` was successfully validated and installed, as evidenced by the "Finished" status and the green checkmark in the deployment dashboard.
- **Rollback Proof:** Earlier deployment failures involving systemd service crashes triggered the automated rollback routine, ensuring the device remained operational.